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Aerospace Engineering

AE-301 AERODYNAMICS

Homework 4

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ANSYS CFD Simulation of a NACA 0012 Airfoil

Introduction

This report presents the setup, execution, and results of a computational fluid dynamics (CFD) simulation of a NACA 0012 airfoil using ANSYS. The objective is to visualize the pressure distribution over the airfoil at a 0 degrees angle of attack and to observe the effects of pressure distribution on this airfoil, gaining insights into the aerodynamic characteristics of this classic symmetric airfoil geometry.

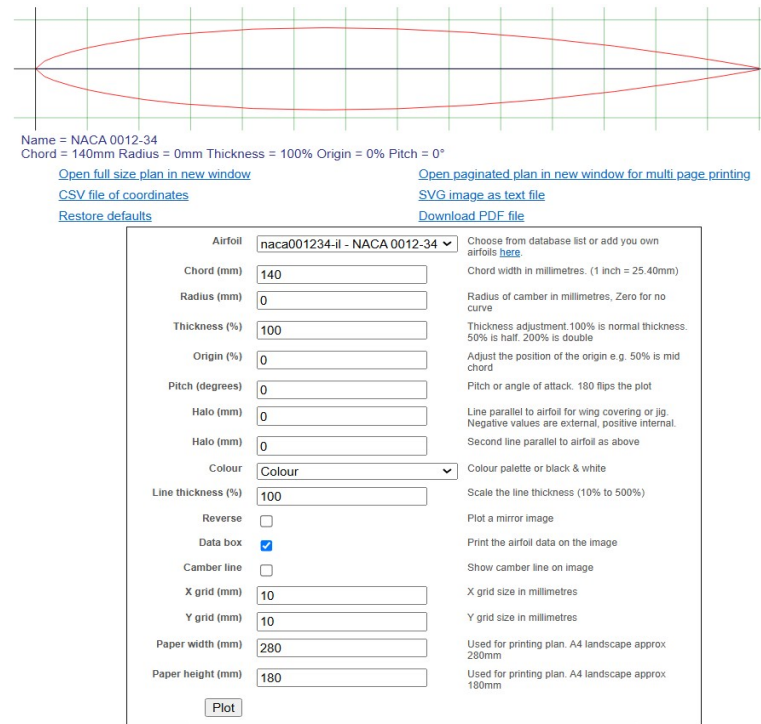
The NACA 0012 airfoil is chosen for its well-known aerodynamic performance and symmetric profile. By performing CFD simulations, it is possible to evaluate flow behaviour, surface pressure distributions, and aerodynamic forces without the need for extensive physical testing.

Geometry and Model Setup

Airfoil Geometry:

- Profile: NACA 0012 (a symmetric airfoil with a maximum thickness of 12% of the chord)
- Chord length: 140 mm
- Span (or width in the out-of-plane direction): 90 mm
- Angle of attack (α): 0°

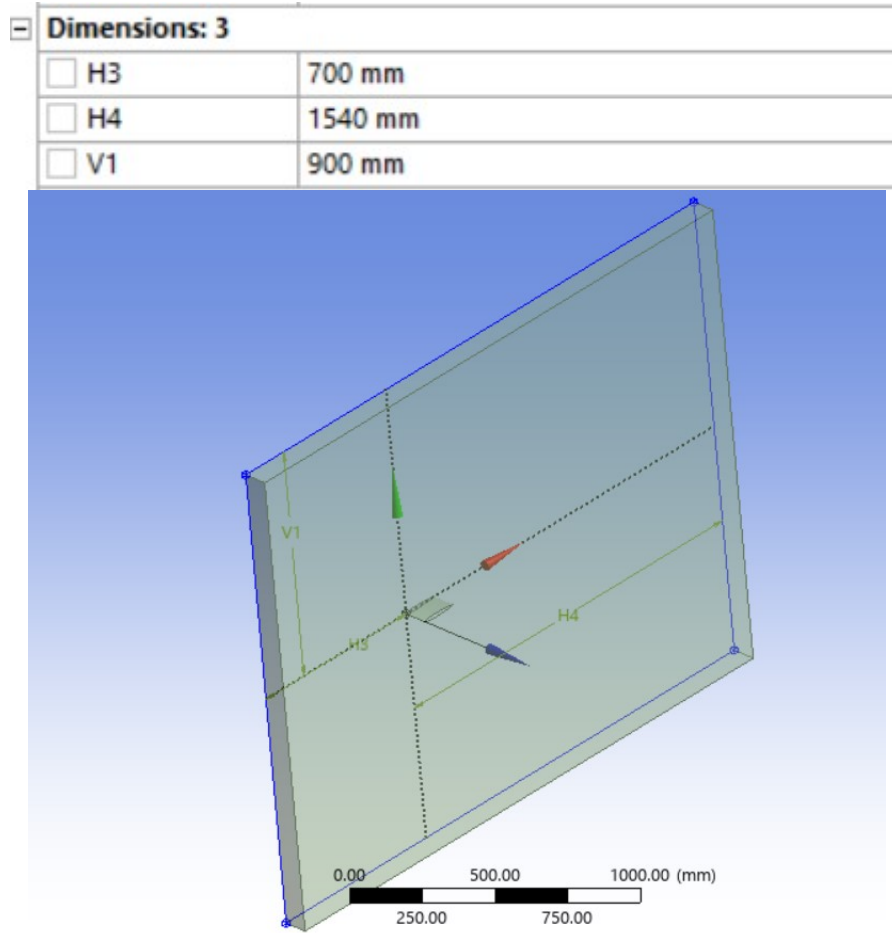
The geometry was created from importing a csv file from an airfoil plotter website where the chord length was adjusted to 140mm. The file was edited on excel to be saved as a table delimited file. It was then generated directly after uploading it within ANSYS DesignModeler.



NACA 0012 plotting data

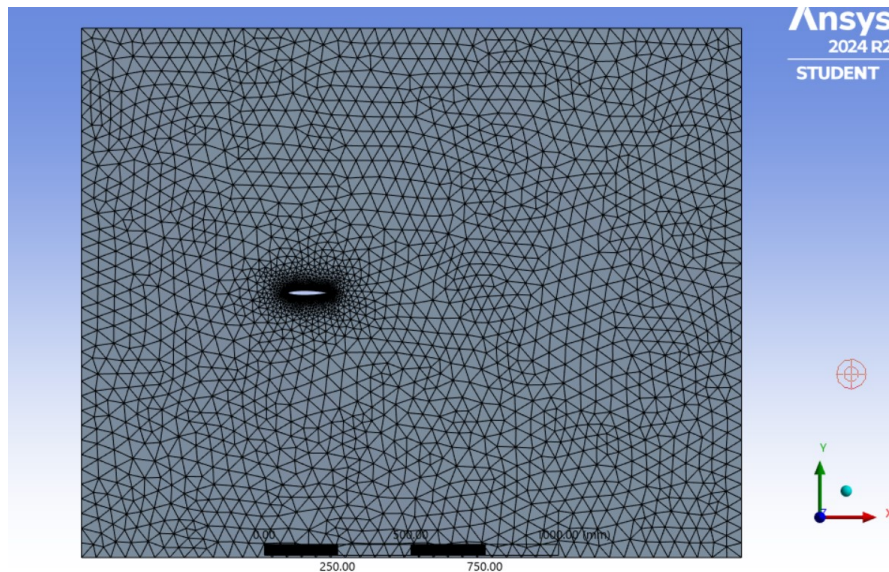
Computational Domain and Boundaries:

A rectangular computational domain was created with the airfoil placed such that the inlet boundary is sufficiently upstream and the outlet boundary is sufficiently downstream of the airfoil to allow the flow to develop and reattach as needed. Typically, the upstream boundary is placed approximately ~5 chord lengths ahead of the airfoil leading edge, and the downstream boundary appr. ~10 chord lengths behind the trailing edge. The top and bottom boundaries are placed far enough away to minimize wall effects.

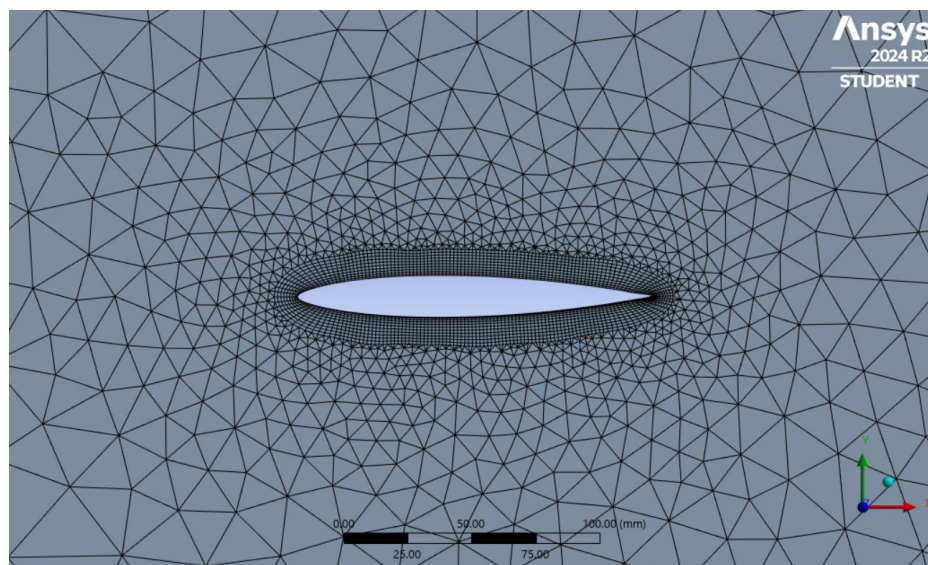


Meshing

For this simulation, a hybrid mesh was generated in ANSYS using a patch-conforming tetrahedral method with inflation layers. The goal was to accurately capture the boundary layer flow near the NACA 0012 airfoil surface, while maintaining adequate resolution and minimizing computational costs.



- **Meshing Method:** A patch-conforming tetrahedral method was selected to ensure good quality elements that conform well to the geometry of the airfoil and the surrounding flow domain.
- **Inflation Layers:** Applied 10 inflation layers at a 1.2 growth rate and a maximum thickness of 16.8 mm ($=0.12c$) to accurately resolve the boundary layer.
- **Face Sizing:** Assigned a 1.4 mm ($=c/100$) element size on a specified face for refined resolution in critical flow regions.
- **Edge Sizing:** Defined 200 divisions along a selected edge to ensure a fine, uniform distribution of elements where high gradient changes are expected.



- **Domain Extent and Symmetry:** The computational domain was sized to minimize the influence of boundary conditions on the flow field around the airfoil. Symmetry planes were defined (sym1 and sym2), distinct boundaries (inlet, outlet and airfoil) were defined to streamline the assignment of boundary conditions and simplify post-processing analysis. reducing the computational domain and taking advantage of symmetric flow conditions.

Physics Setup

Flow Model:

The simulation was run under steady-state, incompressible flow conditions. A suitable turbulence model (k- ω SST) was chosen for its balance of accuracy and stability, especially in predicting flow separation and boundary layer behavior on airfoils.

Boundary Conditions:

- ☞ **Inlet:** Uniform freestream velocity of 19m/s was applied.
- ☞ **Outlet:** Pressure outlet set to atmospheric pressure.
- ☞ **Walls (Airfoil Surface):** No-slip boundary condition with the airfoil surface defined as a smooth wall.
- ☞ **Sym1 and Sym2 Boundaries:** Typically set as symmetry or free-slip walls if the domain is large enough, to minimize boundary layer interference.

Material Properties:

Air was defined as the working fluid with standard conditions (density = 1.225 kg/m³, dynamic viscosity = 1.7894e-5 kg/(m·s) at 15°C).

Solution and Convergence

The simulation was performed using a pressure-based solver. The coupled scheme was selected for pressure-velocity coupling. Second-order spatial discretization was employed for accuracy in all parameters.

Convergence Criteria:

Residuals were monitored to reach e-4. Additionally, lift and drag coefficients were tracked until stable values were obtained, indicating a converged flow field.

After running the calculations, the solution converged after 66 iterations :

```
64 1.0531e-03 2.2938e-07 9.6532e-08 7.6557e-08 1.1272e-05 4.0422e-06 0:02:17 36
65 1.0102e-03 2.2034e-07 9.2628e-08 7.3606e-08 1.0814e-05 3.8415e-06 0:02:15 35
66 9.7031e-04 2.1203e-07 8.8902e-08 7.0819e-08 1.0452e-05 3.6702e-06 0:02:19 34
! 66 solution is converged
Registering ReportDefFiles, ("D:\ansys\kss_HW4_files\dp0\FFF\Fluent\.\cd-rfile.out" "D:\ansys
```

Results and Discussion

Pressure Distribution:

The key output of this simulation is the surface pressure distribution along the airfoil. Pressure contours can be visualized to show high-pressure regions (on the lower surface) and lower-pressure regions (on the upper surface).

Figures and Screenshots:

 **Figure 1:** Pressure contour plot over the airfoil surface.

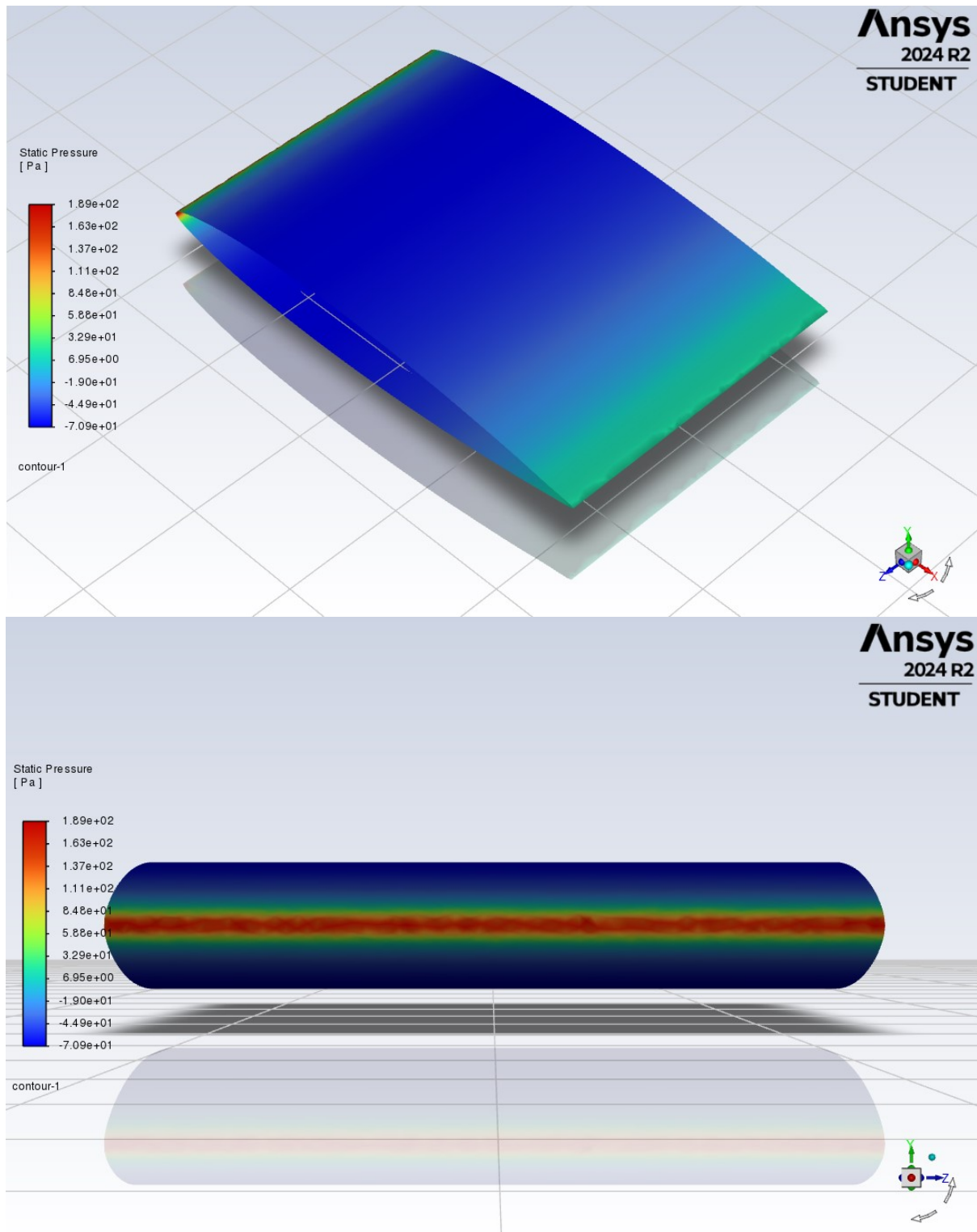
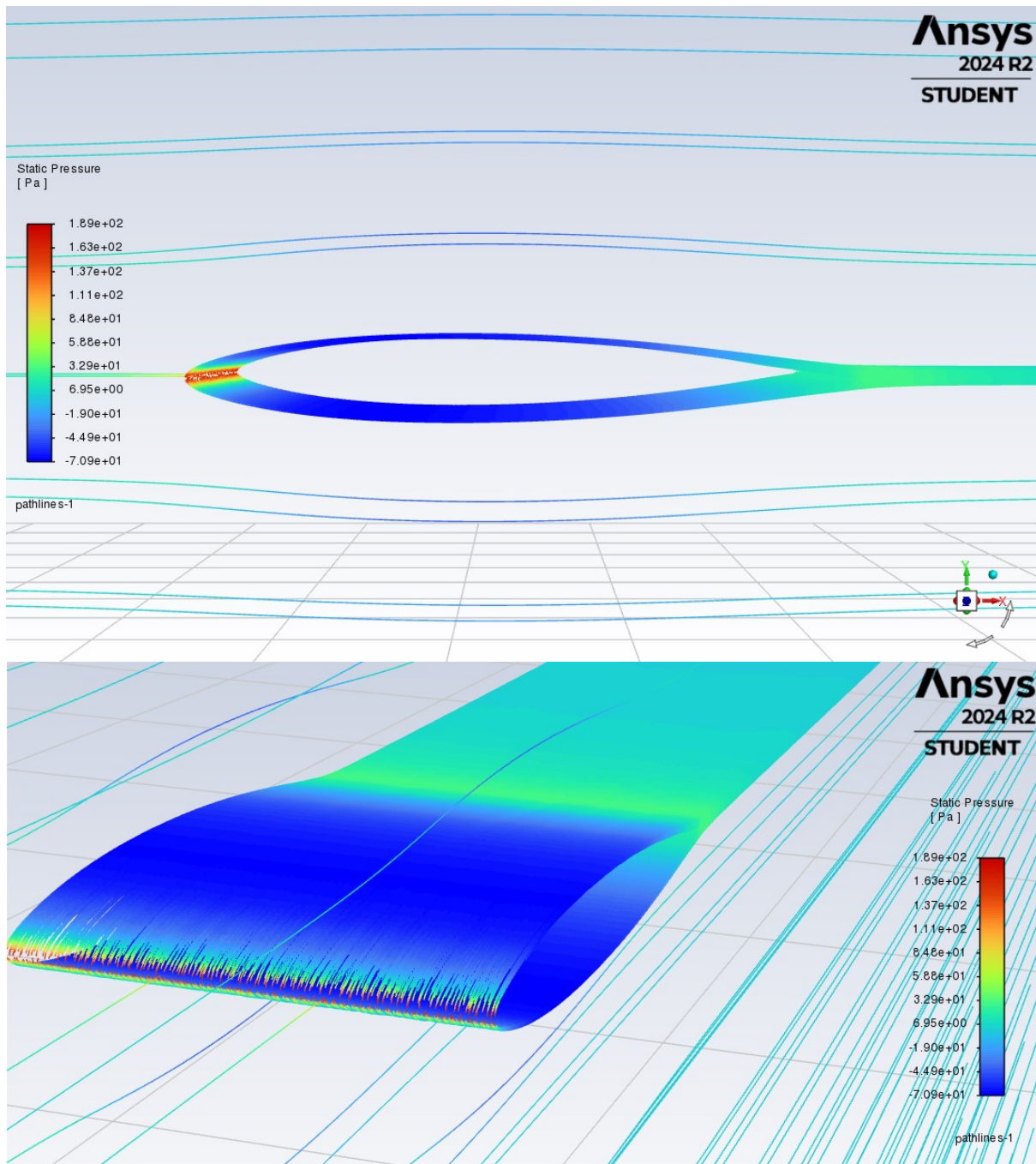


Figure 2: Streamlines showing pressure flow patterns around the airfoil.



With the airfoil set at 0° angle of attack (AoA), the NACA 0012—being a symmetrical airfoil—ideally produces no net lift. These visualizations highlight the pressure distribution around the airfoil being nearly symmetrical about its chord line, with the upper and lower surfaces experiencing similar pressure levels.

Discussion:

In these results, the static pressure contours and pathlines confirm this behavior. At the leading edge, there is a slight increase in pressure due to flow stagnation, and as the air moves around the airfoil, it accelerates symmetrically over both the upper and

lower surfaces. Because the AoA is zero, the flow does not favor one side over the other, preventing the development of a pronounced low-pressure region on either the top or bottom surface. As a result, the integrated pressure forces on the upper and lower surfaces largely cancel out, resulting in negligible net lift.

The pathlines also appear evenly distributed above and below the airfoil, with no significant flow separation or asymmetry. The wake region behind the airfoil remains relatively thin and straight, reflecting minimal lift-induced downwash. These results are consistent with theoretical expectations for a symmetrical airfoil at zero angle of attack, where the flow remains attached and symmetrical, and the aerodynamic coefficients for lift and pitching moment are near zero.

In summary, the simulation results align well with established aerodynamic theory. At 0° AoA, a symmetric NACA 0012 airfoil exhibits nearly symmetric pressure distributions on its upper and lower surfaces, resulting in negligible net lift. The images and contour plots demonstrate stable, uniform flow behavior, confirming the reliability of the simulation setup and the mesh configuration for this particular operating condition.

Conclusion

The CFD simulation of a NACA 0012 airfoil at 0° angle of attack demonstrates flow conditions that closely match theoretical expectations for a symmetric airfoil in aligned freestream conditions. The pressure distribution remains nearly uniform on the upper and lower surfaces, resulting in negligible net lift generation. The pathlines and contour plots confirm that the flow remains attached, symmetrical, and stable around the airfoil, with no significant separation or asymmetry. These findings validate the simulation setup, mesh refinement, and boundary conditions, and provide confidence that the computational approach can accurately capture the essential aerodynamic characteristics of the NACA 0012 profile at this operating condition.

References

- Abbott, I. H., & von Doenhoff, A. E. (1959). *Theory of Wing Sections*. Dover Publications.

- Anderson, J. D. (2010). *Fundamentals of Aerodynamics*. McGraw-Hill.
- Airfoil Plotter website : <http://airfoiltools.com/plotter/index>
- AE301 - Aerodynamics Course Slides.